



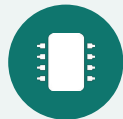
RLHF (PPO)

Humans compare two answers → train a reward model → sample, score and update the LLM. PPO clips each update so the policy moves in small steps.

Feedback: Humans

Reward model: Yes, explicit

Samples while training: Yes (online)



RLAIF

Same pipeline as RLHF, but the preference labels come from an AI judge (e.g. guided by a written constitution). Cheaper and scalable.

Feedback: An AI model

Reward model: Yes, explicit

Samples while training: Yes (online)



DPO

Uses the closed-form optimum to turn the objective into a classification loss on (chosen, rejected) pairs. No RL loop at all.

Feedback: Human or AI pairs

Reward model: No (implicit)

Samples while training: No (offline)



GRPO

Sample a GROUP of answers per prompt; advantage = reward – group mean ($\div$ std). No value network. Popular with verifiable rewards (math, code).

Feedback: RM or a verifier

Reward model: Optional

Samples while training: Yes (online)